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5/5/2026

CSC-121-OA1

Final project

Data Visualization

Asheville Regional Airport

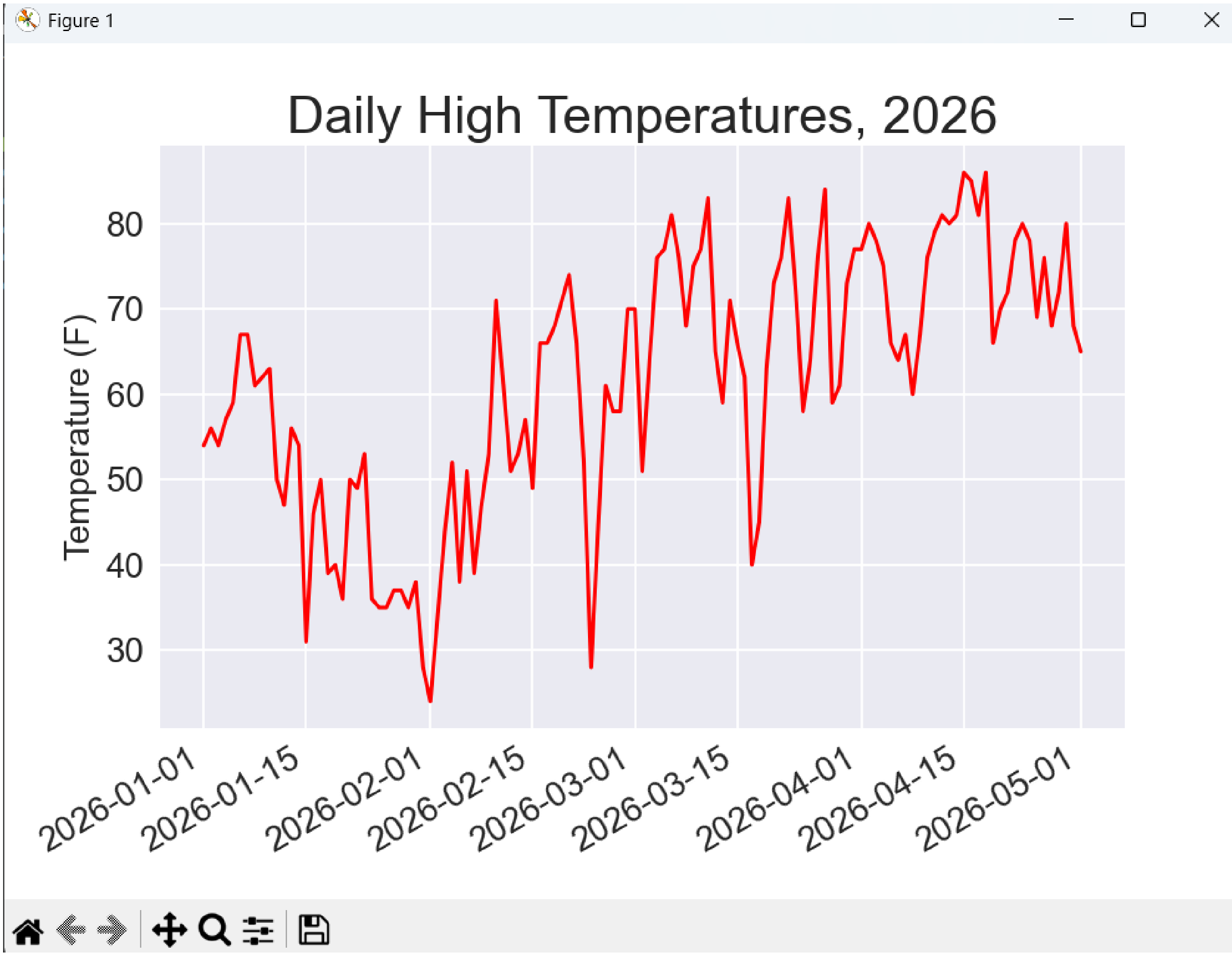
Temperatures for 2026

Print out of the code used for high temperature:

```
from pathlib import Path
import csv
from datetime import datetime

import matplotlib.pyplot as plt

path = Path(__file__).parent / '4306259 (1).csv'
lines = path.read_text().splitlines()

reader = csv.reader(lines)
header_row = next(reader)

# Extract dates and high temperatures.
dates, highs = [], []
for row in reader:
    current_date = datetime.strptime(row[2], '%Y-%m-%d')
    try:
        high = int(row[7])
    except ValueError:
        print(f"Missing data for {current_date}")
```

```
    else:
        dates.append(current_date)
        highs.append(high)

# Plot the high temperatures.
plt.style.use('seaborn-v0_8')
fig, ax = plt.subplots()
ax.plot(dates, highs, color='red')

# Format plot.
ax.set_title("Daily High Temperatures, 2026", fontsize=24)
ax.set_xlabel('', fontsize=16)
fig.autofmt_xdate()
ax.set_ylabel("Temperature (F)", fontsize=16)
ax.tick_params(labelsize=16)

plt.show()
```

Picture of graph:

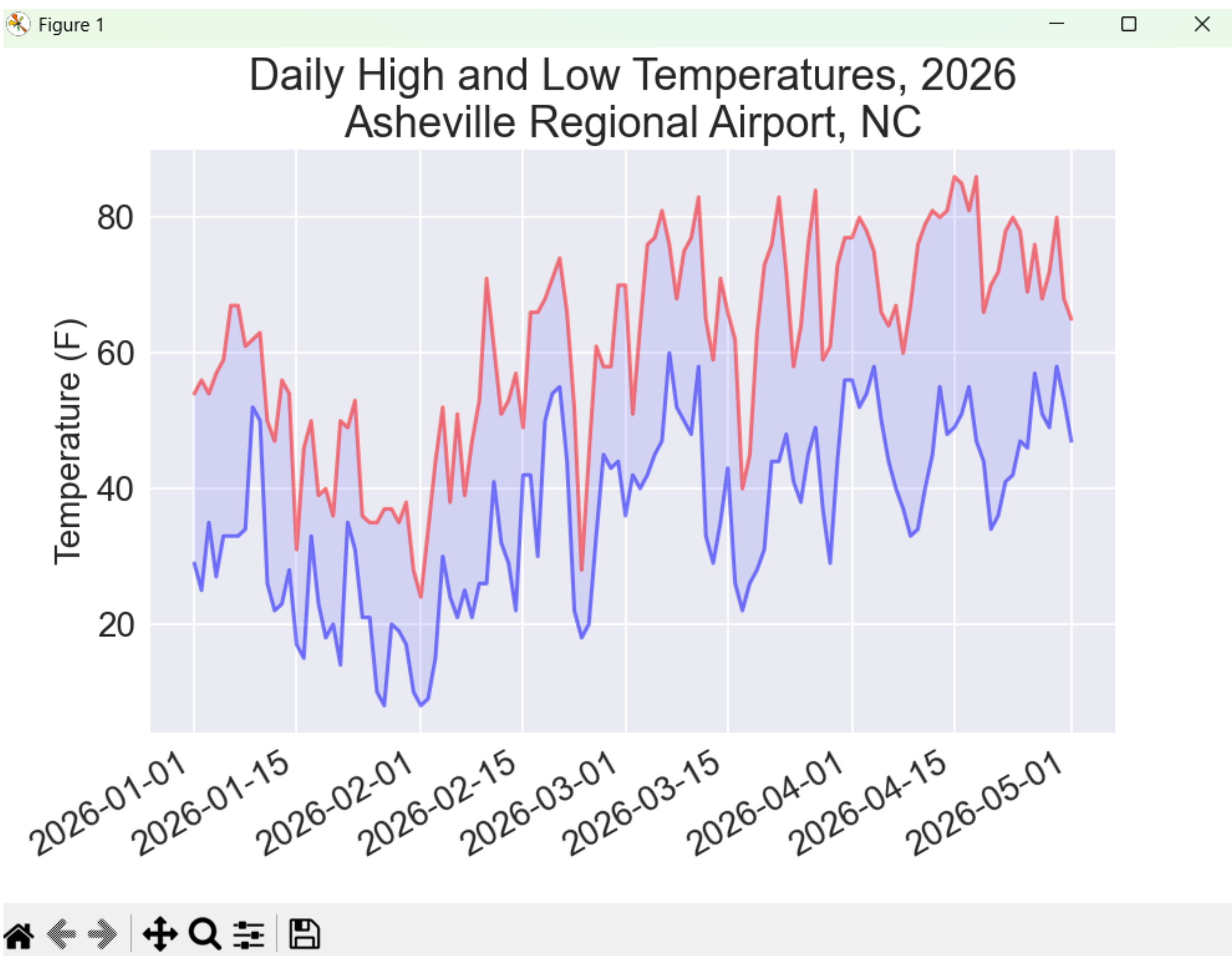


Print out of code used for high and low temperatures:

```
from pathlib import Path
import csv
from datetime import datetime

import matplotlib.pyplot as plt

path = Path(__file__).parent / '4306259 (1).csv'
lines = path.read_text().splitlines()

reader = csv.reader(lines)
header_row = next(reader)

# Extract dates, and high and low temperatures.
dates, highs, lows = [], [], []
for row in reader:
    current_date = datetime.strptime(row[2], '%Y-%m-%d')
    try:
        high = int(row[7])
        low = int(row[8])
    except ValueError:
        print(f"Missing data for {current_date}")
    else:
        dates.append(current_date)
        highs.append(high)
        lows.append(low)

# Plot the high and low temperatures.
plt.style.use('seaborn-v0_8')
fig, ax = plt.subplots()
ax.plot(dates, highs, color='red', alpha=0.5)
```

```
ax.plot(dates, lows, color='blue', alpha=0.5)
ax.fill_between(dates, highs, lows, facecolor='blue', alpha=0.1)

# Format plot.
title = "Daily High and Low Temperatures, 2026\nAsheville Regional Airport, NC"
ax.set_title(title, fontsize=20)
fig.autofmt_xdate()
ax.set_ylabel("Temperature (F)", fontsize=16)
ax.tick_params(labelsize=16)

plt.show()
```

Picture of the graph:



Print out of csv code downloaded for weather data (sample piece):

```
"STATION","NAME","DATE","PRCP","SNOW","SNWD","TAVG","TMAX","TMIN"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-01","0.00","0.0","0.0",,"54","29"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-02","0.00","0.0","0.0",,"56","25"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-03","0.01","0.0","0.0",,"54","35"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-04","0.00","0.0","0.0",,"57","27"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-05","0.00","0.0","0.0",,"59","33"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-06","0.00","0.0","0.0",,"67","33"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-07","0.00","0.0","0.0",,"67","33"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-08","0.00","0.0","0.0",,"61","34"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-09","0.12","0.0","0.0",,"62","52"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-10","2.68","0.0","0.0",,"63","50"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-11","0.00","0.0","0.0",,"50","26"
"USW00003812","ASHEVILLE REGIONAL AIRPORT, NC US","2026-01-12","0.00","0.0","0.0",,"47","22"
```

References:

1. Matthes, Eric (2023). Python Crash Course, 3rd Edition.

No Startch Press, Inc. p300-353.

2. “National Oceanic and Atmospheric Administration” (NOAA)

May 3, 2026

https://www.ncdc.noaa.gov/cdo-web